COLLEGE OF ENGINEERING

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INTRODUCTION TO MACHINE LEARNING

Hospital Length of Stay Prediction

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Abstract

Hospital Length of Stay (LoS) is a critical operational metric in healthcare management. Accurate prediction of LoS enables hospitals to optimize bed allocation, improve patient flow, and reduce operational costs. This study applies machine learning techniques specifically Linear Regression and Random Forest Regression to predict patient LoS using clinical and demographic data sourced from a publicly available hospital dataset comprising 100,000 patient records and 27 features.

Following systematic data preprocessing and exploratory data analysis, both models were trained on an 80/20 train-test split. The Random Forest model achieved a coefficient of determination (R^2) of 0.9410 and a Root Mean Squared Error (RMSE) of 0.5687 days, substantially outperforming the Linear Regression baseline ($R^2 = 0.7522$, RMSE = 1.1661). Feature importance analysis identified prior readmission count, total_issues, and hematological parameters as the strongest predictors of LoS.

These findings demonstrate the viability of ensemble tree-based methods for clinical prediction tasks and provide a foundation for further hyperparameter optimization, cross validation, and deployment in real-world hospital information systems.

Keywords: Hospital Length of Stay, Machine Learning, Random Forest, Linear Regression, Healthcare Analytics, Predictive Modelling

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1. Introduction

1.1 Background

The efficient management of hospital resources is one of the foremost challenges facing modern healthcare systems. Among the many operational variables that affect resource allocation, hospital Length of Stay (LoS) defined as the total number of days a patient remains admitted from the date of admission to the date of discharge stands out as particularly consequential. An inaccurately predicted LoS can lead to bed shortages, delayed elective procedures, increased healthcare costs, and diminished quality of patient care (Daghistani et al., 2019).

In recent years, the widespread adoption of Electronic Health Records (EHRs) and Hospital Information Systems has made large volumes of structured clinical data available for computational analysis. Machine learning (ML) methods have emerged as powerful tools for extracting predictive signals from such high-dimensional datasets, offering the potential to substantially improve upon traditional statistical approaches.

1.2 Problem Statement

Despite the availability of rich clinical data, many hospitals still rely on heuristic estimates or simple regression models for LoS prediction. These approaches fail to capture the complex, non-linear interactions among clinical variables. This project addresses the need for accurate, data-driven LoS predictions by comparing a linear baseline model with a non-parametric ensemble method.

1.3 Objectives

The specific objectives of this study are:

- To perform exploratory data analysis (EDA) and data quality assessment on a hospital patient dataset.
- To preprocess and engineer features suitable for machine learning models.
- To train and evaluate a Linear Regression and a Random Forest Regression model.
- To compare model performance using R^2 and RMSE metrics.
- To identify the clinical variables most predictive of hospital LoS.

1.4 Scope and Limitations

This study is confined to the publicly available Microsoft R Server Hospital LoS dataset. The models are trained as batch predictors and are not evaluated in a real-time clinical setting. Temporal information (admission date, discharge date) was excluded from modelling to avoid data leakage, and facility-level effects were not incorporated in the primary analysis.

2. Literature Review

2.1 Traditional Approaches to LoS Prediction

Early approaches to hospital LoS prediction relied on administrative groupings such as Diagnosis-Related Groups (DRGs) and generalised linear models (GLMs). While computationally tractable, these methods frequently exhibited poor predictive accuracy due to their inability to model interactions among clinical variables (Tsai et al., 2016). Multivariate regression approaches improved upon these baselines but remained constrained by assumptions of linearity and normality of residuals.

2.2 Machine Learning Applications in Healthcare

The application of machine learning to clinical prediction tasks has grown substantially over the past decade. Decision tree ensembles particularly Random Forests (Breiman, 2001) and Gradient Boosted Trees have demonstrated strong performance across a range of clinical prediction tasks, including mortality prediction, readmission risk scoring, and sepsis detection. Their ability to model non-linear relationships and feature interactions without requiring explicit feature engineering makes them well-suited for heterogeneous clinical datasets.

2.3 LoS Prediction Using Machine Learning

Several prior studies have explored ML-based LoS prediction. Turgeman et al. (2017) applied Random Forest and CART models to ICU patient data, reporting R^2 values above 0.80 for ensemble methods. Barnes et al. (2016) demonstrated that incorporating readmission count and comorbidity indicators markedly improved prediction accuracy. Harutyunyan et al. (2019) proposed multitask learning frameworks for clinical time-series data, achieving further improvements by jointly predicting LoS alongside mortality risk.

Collectively, this literature establishes a strong prior that ensemble tree methods consistently outperform linear baselines for LoS prediction, and that readmission count and comorbidity burden are robust predictors findings consistent with the present study.

3. Dataset Description

3.1 Source

The dataset used in this study was sourced from the Microsoft R Server Hospital Length of Stay dataset, publicly available at https://microsoft.github.io/r-server-hospital-length-of-stay/input_data.html. It is records from multiple healthcare facilities.

3.2 Dataset Overview

The dataset comprises 100,000 patient records and 28 columns (features plus target). There are no missing values in the raw data, and no duplicate records were identified. One erroneous negative glucose value was detected and corrected during preprocessing.

Property	Value
Total Records	100,000
Total Features (raw)	28
Target Variable	lengthofstay (days)
Missing Values	0 (raw); 1 corrected (glucose < 0)
Duplicate Rows	0
Data Type	Mixed (continuous, binary, categorical)

Table 1: Dataset overview statistics.

3.3 Feature Descriptions

The 28 original features span four categories:

3.3.1 Administrative Features (dropped prior to modelling)

Feature	Type	Description
eid	Integer	Unique patient encounter identifier
vdate	Date	Date of hospital admission
discharged	Date	Date of patient discharge
facid	Categorical	Facility identifier (A–E)

Table 2: Administrative features excluded from modelling.

3.3.2 Demographic and Clinical History Features

Feature	Type	Description
rcount	Ordinal (0–5+)	Prior inpatient readmission count
gender	Binary	Patient gender (F / M)
secondarydiagnosisnonicd9	Integer (0–10)	Number of secondary diagnoses

Table 3: Demographic and clinical history features.

3.3.3 Comorbidity Indicators (Binary Flags)

Eleven binary comorbidity flags indicate the presence (1) or absence (0) of specific conditions:

Feature	Condition
dialysisrenalendstage	Dialysis / Renal End Stage Disease
asthma	Asthma
irondef	Iron Deficiency
pneum	Pneumonia
substancedependence	Substance Dependence
psychologicaldisordermajor	Major Psychological Disorder
depress	Depression
psychother	Other Psychological Disorder
fibrosisandother	Fibrosis and Other Conditions
malnutrition	Malnutrition
hemo	Haemoglobin Disorder

Table 4: Binary comorbidity indicator features.

3.3.4 Continuous Clinical Measurements

Feature	Unit / Range	Description
hematocrit	% (4.4–24.1)	Red blood cell volume fraction
neutrophils	% (0.1–245.9)	White blood cell neutrophil count
sodium	mEq/L (124.9–151.4)	Serum sodium concentration
glucose	mg/dL (corrected)	Blood glucose level
bloodureanitro	mg/dL (1–682.5)	Blood urea nitrogen
creatinine	mg/dL (0.22–2.04)	Serum creatinine
bmi	kg/m² (22.0–38.9)	Body Mass Index
pulse	bpm (21–130)	Heart rate
respiration	breaths/min (0.2–10.0)	Respiratory rate

Table 5: Continuous clinical measurement features.

3.4 Target Variable

The target variable, lengthofstay, is a discrete integer representing the number of inpatient days. Summary statistics are presented below:

Statistic	Value
Minimum	1 day
25th Percentile	2 days
Median	4 days
Mean	4.00 days
75th Percentile	6 days
Maximum	17 days
Standard Deviation	2.36 days

Table 6: Descriptive statistics for the target variable (lengthofstay).

4. Exploratory Data Analysis

4.1 Data Quality Assessment

Initial data quality checks confirmed the following:

- No missing values were present in any column after loading.
- No duplicate patient records were identified across the 100,000 rows.
- One record contained a physiologically impossible glucose value of −1.006 mg/dL, which was identified as a data entry error and corrected by imputation with the column median.
- The rcount column contained a categorical label '5+' for patients with five or more prior readmissions, which was converted to the integer value 5 for numerical processing.

4.2 Target Variable Distribution

The target variable (lengthofstay) ranges from 1 to 17 days, with a mean of 4.00 days and a right-skewed distribution. Approximately 50% of patients were discharged within 4 days, while the upper quartile extends to 6 days. The right tail represents complex cases with extended inpatient needs.

4.3 Correlation Analysis

Pearson correlation coefficients between each feature and the target variable were computed. The strongest positive correlations with lengthofstay are summarized below:

Feature	Correlation with LoS	Direction
rcount (readmission count)	0.7495	Positive
psychologicaldisordermajor	0.2867	Positive
hemo (haemoglobin disorder)	0.2177	Positive
irondef (iron deficiency)	0.1938	Positive
psychother	0.1917	Positive
malnutrition	0.1744	Positive
dialysisrenalendstage	0.1697	Positive
bloodureanitro	0.1483	Positive
substancedependence	0.1479	Positive
hematocrit	-0.0640	Negative

Table 7: Top correlations between features and the target variable (lengthofstay).

Notably, readmission count (rcount) exhibits a strong linear correlation of 0.75, indicating it is by far the most informative single predictor. Most comorbidity indicators show moderate positive correlations, consistent with clinical expectations that patients with multiple conditions require longer hospital stays.

4.4 Categorical Feature Analysis

The categorical variable gender was encoded as binary (Female = 0, Male = 1) and showed a weak positive correlation with LoS (r = 0.07), suggesting a marginal tendency for male patients to have slightly longer stays. The facility identifier (facid) with five unique values (A–E) was excluded from the modelling phase, though it may carry predictive signal warranting future investigation through target encoding.

5. Methodology

5.1 Data Preprocessing Pipeline

5.1.1 Outlier Correction

A single record with a glucose value below zero was identified as an outlier inconsistent with physiological norms. The value was set to NaN and subsequently imputed using the column median (≈ 142.09 mg/dL), a robust imputation strategy appropriate for data with potential outliers.

5.1.2 Feature Removal

Four columns were removed from the feature set prior to modelling:

- eid - a non-informative unique identifier.

- vdate - admission date; temporal features risk leakage in a static model.
- discharged - discharge date; directly encodes LoS and would cause leakage.
- facid - facility ID; excluded in the primary analysis (future work).

5.1.3 Encoding

The ordinal feature rcount ('0', '1', '2', '3', '4', '5+') was converted to integer representation (0–5). The binary categorical feature gender ('F', 'M') was label-encoded to (0, 1). All remaining features were already numeric.

5.1.4 Feature Engineering

A composite feature, `total_issues`, was engineered by summing the 11 binary comorbidity indicator columns for each patient. This single integer (range 0–11) captures the overall comorbidity burden per patient, consolidating correlated binary flags into one informative signal. The resulting dataset contains 24 input features after this addition.

5.2 Train-Test Split

The 100,000-record dataset was partitioned into a training set (80%, n = 80,000) and a holdout test set (20%, n = 20,000) using stratified random sampling with a fixed random seed (random_state = 42) to ensure reproducibility.

5.3 Feature Scaling

For the Linear Regression model, features were standardized using StandardScaler (zero mean, unit variance). Scaling was fitted exclusively on the training set and applied to the test set to prevent data leakage. Random Forest models are invariant to monotonic feature transformations; therefore, unscaled features were used for that model.

5.4 Models

5.4.1 Linear Regression

Ordinary Least Squares (OLS) linear regression was implemented using scikit-learn’s LinearRegression class. This model serves as the interpretable baseline against which the ensemble method is compared. It assumes a linear additive relationship between features and the target:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

5.4.2 Random Forest Regression

Random Forest Regression (Breiman, 2001) is an ensemble method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees. Key hyperparameters used were n_estimators = 100 (number of trees) and random_state = 42. The model inherently handles non-linear relationships, feature interactions, and is robust to outliers and irrelevant features.

5.5 Evaluation Metrics

Two standard regression evaluation metrics were used:

- R^2 (Coefficient of Determination): Proportion of variance in LoS explained by the model. Values closer to 1.0 indicate better fit.
- RMSE (Root Mean Squared Error): Average prediction error in the original unit (days). Lower values indicate higher accuracy.

6. Results

6.1 Model Performance Comparison

Both models were evaluated on the 20,000-record holdout test set. Results are summarized below.

Model	R^2 Score	RMSE (days)	Training Features
Linear Regression	0.7522	1.1661	Scaled (StandardScaler)
Random Forest (100 trees)	0.9410	0.5687	Unscaled

Table 8: Performance comparison of both models on the holdout test set.

The Random Forest model substantially outperformed Linear Regression across both metrics. It explained 94.10% of the variance in LoS (vs. 75.22% for Linear Regression) and achieved an average prediction error of only 0.5687 days a 51% reduction in RMSE compared to the linear baseline. These results are consistent with the expectation that LoS prediction involves complex non-linear relationships that tree-based ensembles are better equipped to capture.

6.2 Feature Importance Analysis

Random Forest's built-in feature importance scores (mean decrease in impurity) were computed across all 100 trees. The top 10 most influential features are presented in the table below.

Rank	Feature	Importance Score	Category
1	rcount	0.5643	Clinical History
2	total_issues	0.2132	Engineered Feature
3	hematocrit	0.0387	Lab Value
4	bmi	0.0255	Anthropometric
5	glucose	0.0255	Lab Value
6	creatinine	0.0253	Lab Value
7	sodium	0.0251	Lab Value

Rank	Feature	Importance Score	Category
8	respiration	0.0247	Vital Sign
9	pulse	0.0238	Vital Sign
10	neutrophils	0.0162	Lab Value

Table 9: Top 10 Random Forest feature importance scores.

The readmission count (rcount) dominates all other features with an importance score of 0.5643, accounting for over 56% of the model's predictive power. This strongly corroborates the correlation analysis and aligns with clinical literature suggesting that patients with frequent prior hospitalizations typically present with more complex conditions requiring longer treatment.

The engineered feature `total_issues` ranked second with an importance score of 0.2132, confirming that aggregated comorbidity burden is a powerful predictor of extended stay.

6.3 Linear Regression Coefficients

While full coefficient values are not tabulated here, the directional signs of the linear model's coefficients aligned with clinical expectations: readmission count, comorbidity flags, and abnormal lab values all contributed positively to predicted LoS, while higher hematocrit (an indicator of hematological health) contributed negatively, consistent with the observed negative correlation ($r = -0.064$).

7. Discussion

7.1 Interpretation of Results

The results confirm that Random Forest Regression is a significantly more effective model for hospital LoS prediction than linear regression. The linear model's R^2 of 0.7522, while respectable for a baseline, leaves approximately 25% of variance unexplained attributable to non-linear feature interactions that OLS cannot capture. The Random Forest's R^2 of 0.9410 suggests that the ensemble of decision trees successfully models these interactions.

The dominance of rcount in the feature importance ranking suggests that prior healthcare utilization is a strong structural predictor of current stay duration. This has practical implications: patients identified as high readmitters at the point of admission may benefit from targeted care management programs designed to address root causes and reduce LoS.

7.2 Clinical Relevance

From a clinical perspective, the identified predictors align well with established risk stratification frameworks. Comorbidity burden, nutritional status, and renal function are standard components of early warning scoring systems. An ML model capable of integrating all these signals simultaneously offers a data-driven complement to clinical judgement, potentially flagging high-risk patients earlier in the admission process.

7.3 Limitations

Several limitations of the current study should be acknowledged:

- Temporal features (admission date, discharge date) were excluded to prevent leakage, but seasonal and day-of-week effects on LoS are well-documented in the clinical literature.
- The facility identifier (facid) was dropped, potentially discarding facility-level variation that could improve model accuracy.
- The engineered `total_issues` feature, while informative, aggregates conditions with potentially different clinical weights equally. Future work could apply weighted scoring (e.g., Charlson Comorbidity Index) instead of a simple count.
- Model evaluation was performed on a single train-test split rather than k-fold cross-validation, which may introduce variance in performance estimates.
- The binary encoding of gender assumes a simple ordinal relationship, which may not adequately represent the feature's relationship with LoS.

7.4 Future Work

Future extensions of this work should consider:

- Applying k-fold cross-validation ($k = 5$ or 10) to obtain more robust performance estimates.
- Hyperparameter tuning of the Random Forest model using GridSearchCV or RandomizedSearchCV.
- Comparing additional algorithms such as Gradient Boosting (XGBoost, LightGBM).

8. Conclusion

This study presented a machine learning pipeline for predicting hospital Length of Stay using a dataset of 100,000 patient records. Following data preprocessing including outlier correction, feature encoding, and feature selection, two regression models were trained and evaluated.

The Random Forest Regression model achieved an R^2 of 0.9410 and RMSE of 0.5687 days, substantially outperforming the Linear Regression baseline ($R^2 = 0.7522$, RMSE = 1.1661 days). Feature importance analysis confirmed that readmission count is by far the most predictive feature, followed by psychological comorbidities and hematological indicators.

These results highlight the value of ensemble machine learning methods for clinical prediction tasks and provide a solid methodological foundation for developing decision-support tools in hospital settings. With further validation on real-world EHR data and appropriate regulatory review, models of this type have the potential to meaningfully improve resource planning, patient flow, and care quality in inpatient healthcare environments.

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